

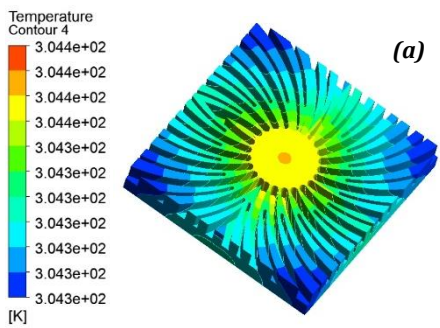
Evaluating Heat Sink Performance

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Motivation: One of the most awaited CFD simulations that I have been waiting to perform was the thermal analysis of heat sinks for companies that design and manufacture heat sinks, especially for computer systems for the effective cooling of integrated circuits (computer chips). My experience showed me that computer modelling is paramount in providing accurate data and simulation models for thermal analysis and fluid dynamics around a body with known boundary conditions. They also help us to validate the results obtained manually.

Project Description: For this project, IBM has given our CFD lab a contract to perform thermal analysis on a novel heat sink and an existing heat sink design provided to us by IBM. They have asked us to compare the conventional and market-available heat sinks in terms of their pin fin efficiency. We have considered the two heat sink designs to have similar fin lengths to ensure a fair comparison while also ensuring similar fabrication techniques and assembly costs. IBM has also requested us to consider a maximum computer chip heat generation of 50 W which decreases radially from the center to the boundaries. We have been asked to demonstrate the use of optimal computational mesh in the modelling of heat sinks. Furthermore, we have also been asked to validate the software's capability to simulate heat generation and the contact between two different objects.

Testing: We started by modelling the novel heat sink provided to us by IBM for different mesh element sizes such as 1.1 mm, 0.8 mm, and 0.55 mm. We then performed the thermal analysis using UDF for the heat source variation along with other boundary conditions. When we plotted a graph of the temperature vs y-coordinates for the different mesh element sizes and found the difference between the mesh efficiency to be very minimal. We can thus determine the optimal mesh sizing to be 0.8mm in this case from the graph. We then performed the same analysis on the market available heat sink in SpaceClaim and performed the meshing operation with different element sizes such as 1 mm, 0.7 mm, and 0.8 mm. We also calculated the fin efficiency to be 83% in the case of the IBM novel heat sink design and approximately 78% in the case of the market-available heat sink design. Thus, we found the variation to be almost the same for all three mesh sizes in this case as shown in the graph. Finally, we performed the validation of the heat generation in a solid object to be equal to 312 K and verified it manually with a difference of only 0.1 K. We also performed the validation of contact between the two thermally interacting objects to be equal to approximately 298.55 K experimentally and manually obtaining a value of 298.45 K again with a difference of only 0.1 K. Thus, validating our entire simulation and successfully verifying the accuracy of the project.



(a) We started by modelling the novel heat sink provided to us by IBM for different mesh element sizes such as 1.1 mm, 0.8 mm, and 0.55 mm. We then performed the thermal analysis using UDF for the heat source variation along with other boundary conditions. When we plotted a graph of the temperature vs y-coordinates for the different mesh element sizes and found the difference between the mesh efficiency to be very minimal. We can thus determine the optimal mesh sizing to be 0.8mm in this case from the graph. We then performed the same analysis on the market available heat sink in SpaceClaim and performed the meshing operation with different element sizes such as 1 mm, 0.7 mm, and 0.8 mm. We also calculated the fin efficiency to be 83% in the case of the IBM novel heat sink design and approximately 78% in the case of the market-available heat sink design. Thus, we found the variation to be almost the same for all three mesh sizes in this case as shown in the graph. Finally, we performed the validation of the heat generation in a solid object to be equal to 312 K and verified it manually with a difference of only 0.1 K. We also performed the validation of contact between the two thermally interacting objects to be equal to approximately 298.55 K experimentally and manually obtaining a value of 298.45 K again with a difference of only 0.1 K. Thus, validating our entire simulation and successfully verifying the accuracy of the project.

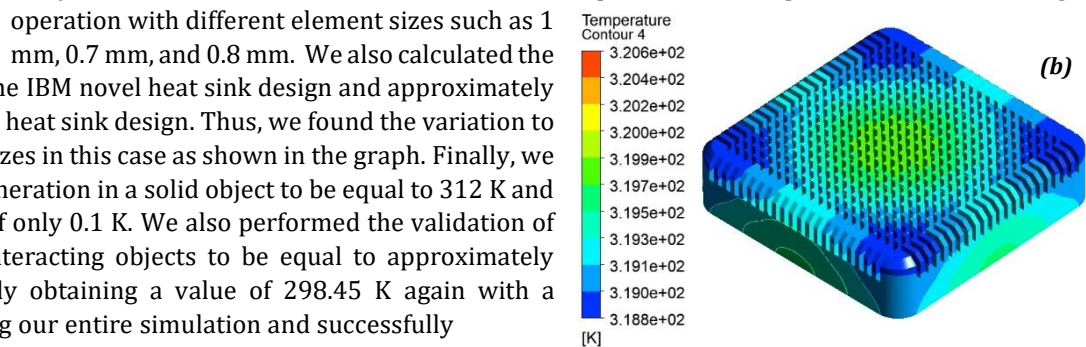
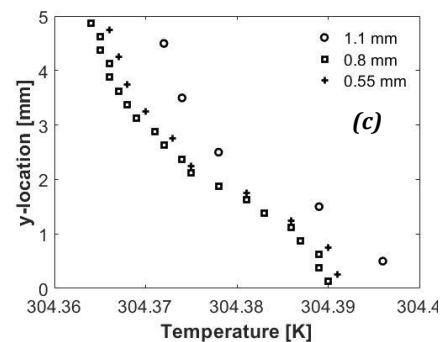
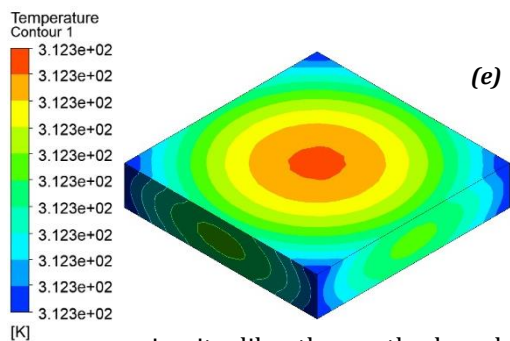


Figure: (a) IBM Novel Heat Sink (b) Market Heat Sink (c), (d) Graphs – Mesh sensitivity of novel heat sink and market heat sink respectively (e) Validation of heat generation (f) Validation of contact between two thermally interacting objects

Challenges: The most challenging part of the project was to try and understand how to efficiently perform the entire thermal analysis of the heat sink and compare the two types of heat sink designs. Performing multiple mesh sensitivity analyses with different element sizes was an added burden due to its time-consuming nature.



(e) Meaning: Now that I have performed the heat transfer simulation, I can see many applications of it in the industry and academia using computer modelling. For example, I can apply this idea to simulate the thermal analysis of heat exchangers such as high-speed computer fans which are used to effectively cool the integrated

circuits like the motherboard inside a computer, enabling the optimal performance of the system. But this obviously would be more complicated with a much deeper understanding of the boundary conditions and better skills with the meshing of certain complex geometries.

